GNN-Augmented BP Architectures for QLDPC Decoding

Jothiradithya Sai Konduru[1], Anish Chedalla[1], and Nithin Raveendran[2]

[1]Paradise Valley High School, Phoenix, AZ, USA
[2]Department of Electrical and Computer Engineering, University of Arizona, Tucson, AZ, USA

Bivariate bicycle (BB) quantum low-density parity-check (QLDPC) codes combine finite encoding rate with low-weight checks, but their loopy Tanner graphs remain challenging for iterative decoding. We study a Tanner-graph neural network (TannerGNN) that adds qubit-wise log-likelihood corrections to belief propagation (BP) under biased CSS noise. The network is trained with a per-qubit focal loss on the sampled error, and its checkpoint is selected with 10-iteration flooding BP. On the $[\![72, 12, 6]\!]$ BB code at $p = 0.04$ and $\eta = 20$, it reduces the logical error rate (LER) of that decoder by $12.5\% \pm 0.6\%$ on 20,000 held-out syndromes (five training seeds; paired McNemar $p < 10^{-36}$ for every seed). The gain does not transfer to a stronger reference decoder, serial min-sum BP-OSD: the correction never improves it and, after Holm correction, is significantly worse for 11 of the 15 seed and error-rate pairs with $p \geq 0.03$. Most BP-OSD failures are also BP failures, the GNN mainly repairs shots on which BP converges slowly and which BP-OSD already decodes, and weight-three errors that cover half of a stabilizer defeat BP with or without the GNN but never BP-OSD. Baseline experiments show that BP schedule, update rule, iteration count and OSD settings change the LER by orders of magnitude, and we prove that uniformly rescaling the channel LLRs cannot change the hard decisions of unclipped min-sum BP. Revealing the true per-qubit error rates, by contrast, reduces the BP-OSD LER by up to 88% under synthetic heterogeneous noise and by 55% under circuit-level noise.

Nithin Raveendran: nithin@arizona.edu

1 Introduction

Bivariate bicycle (BB) codes provide a concrete finite-length setting in which QLDPC codes combine non-vanishing encoding rate, weight-six stabilizer checks, and syndrome-extraction circuits compatible with low-degree connectivity [1]. IBM's current fault-tolerant architecture is explicitly based on BB codes, making the finite-length decoding behavior of this family practically relevant as well as theoretically interesting [2, 3]. The canonical examples include the $[\![72, 12, 6]\!]$, $[\![144, 12, 12]\!]$, and $[\![288, 12, 18]\!]$ codes studied here.

The decoding bottleneck is not the absence of efficient iterative algorithms, but their finite-length failure modes. BP is attractive because its message updates are local and parallelizable, yet short cycles and degeneracy in QLDPC Tanner graphs create correlated messages and non-convergent or incorrect fixed points [4]. OSD post-processing can recover many BP failures, while recent learned decoders use graph neural networks (GNNs) either to replace BP or to intervene between BP stages [5, 6]. These approaches make the choice of baseline unusually consequential: a neural correction developed for one BP schedule or stopping rule need not address the residual failures of a stronger decoder.

A second issue is adaptivity. Hardware noise is neither spatially nor temporally uniform, and calibration information can therefore be useful to a decoder. However, three distinct sources of information should not be conflated: a global estimate of the physical error rate, spatially resolved per-qubit calibration data, and syndrome history. A decoder may respond differently to each. In particular, changing the magnitude of a uniform channel prior has a restricted effect on min-sum BP, whereas per-qubit priors can change the relative reliability ordering that drives both BP and OSD.

We study a GNN-augmented BP pipeline with

two questions in view: *what gain does the learned correction provide relative to the BP decoder it is developed with, and what additional information is needed for noise adaptation?* We compare decoder configurations, training and evaluation baselines, and different forms of calibration information to isolate the source of the observed performance changes.

1.1 Related work

Quantum Tanner graphs combine short cycles with degeneracy, and both cause non-convergence and finite-length error floors in BP [4, 7, 8]. Trapping-set analysis makes these failures structural rather than mysterious: specific subgraphs, including symmetric configurations created by stabilizers, can trap iterative decoders [9], and degeneracy itself can be exploited inside BP [10].

Several families of methods repair such failures, with different latency and complexity tradeoffs. OSD post-processing [4, 11, 12] is the most widely used reference. Alternatives include stabilizer inactivation [13], check-agnosia [14], localized statistics decoding [15], syndrome-based linear programming [16], collective bit flipping [17], and trapping-set-aware iterative decoders of linear complexity [18]. For BB codes in particular, recent work targets real-time decoding: Relay-BP [19], graph rewiring with ensembles of normalized min-sum decoders [20], diversity and ensemble methods [21], ambiguity clustering [22], and hardware implementations of BP-OSD [23].

Learned decoders either modify BP messages or take over part of the inference. Neural BP with a degeneracy-aware loss [24], GNN stages interleaved with BP [5], learned message passing [6], circuit-level neural decoders for BB codes [25], and calibration-conditioned decoders [26] report both strong gains and limits in scaling and transfer. In this landscape serial min-sum BP-OSD is a useful, fully specified reference decoder, not the state of the art, and we use it only in that role. Our question is narrower: what a learned LLR correction gains relative to the BP configuration it is developed with, whether that gain survives a stronger reference decoder, and what side information adaptation actually needs.

1.2 Contributions

Our main findings are:

1. **A held-out GNN gain over the BP decoder it is developed with.** On $[\![72, 12, 6]\!]$ at $p = 0.04$ and $\eta = 20$, TannerGNN+flooding BP reduces LER by $12.5\% \pm 0.6\%$ on 20,000 held-out syndromes over five training seeds (McNemar $p < 10^{-36}$ for every seed). The training rates span $p \in [0.005, 0.045]$; at $p = 0.06$, outside that range, the reduction is $10.6\% \pm 0.7\%$. The whole gain is in the Z component; adding the correction to the X component as well only adds failures there (Sec. 5.3).

2. **Baseline configuration and alignment can dominate the apparent learned gain.** Serial min-sum BP with OSD post-processing is substantially stronger than the flooding BP used to select the GNN checkpoint. The same learned correction never improves this stronger decoder and, after Holm correction, is significantly worse for 11 of the 15 seed and error-rate pairs with $p \geq 0.03$. A paired failure-set analysis shows that 93% of BP-OSD failures are also BP failures (61% the other way) and that most shots the GNN repairs are already decoded by BP-OSD. Separate experiments on $[\![288, 12, 18]\!]$ show order-of-magnitude sensitivity to BP schedule, update rule, iteration count, and OSD settings. The decoder used for model selection and the evaluation decoder are therefore part of the experimental condition rather than interchangeable implementations of "BP" (Secs. 5.1 and 5.4).

3. **A recurring BP failure that the GNN does not repair.** When a weight-three Z error covers half of a weight-six stabilizer, the error and its stabilizer complement are equally likely and logically equivalent. BP, 100-iteration BP and GNN-BP fail on every such shot in our test sets and BP-OSD on none; the GNN's bitwise loss gives both halves the same target, so it cannot learn to break the tie (Secs. 3.2 and 5.4).

4. **Global prior scaling and spatial calibration carry different information.** We prove that hard decisions of unclipped min-sum BP are invariant to a uniform positive rescaling of the channel LLRs within a CSS component. This result limits adaptivity obtained only by changing a uniform BP prior; it does not rule out a learned network using a global noise scalar as context. In synthetic heterogeneous-noise experiments, providing the true per-

qubit rates to BP-OSD yields large oracle improvements, motivating spatially resolved calibration inputs. Because code size and operating point co-vary in these experiments, we do not attribute the cross-code trend to distance alone (Secs. 5.2 and 5.5).

2 Background

2.1 CSS codes and bivariate bicycle construction

A CSS code [27, 28] is defined by matrices $\mathbf{H}_x, \mathbf{H}_z \in \mathbb{F}_2^{m\times n}$ with $\mathbf{H}_x\mathbf{H}_z^\top = \mathbf{0}$. The syndrome of an error (e_x, e_z) decomposes as $s_x = \mathbf{H}_x e_z$, $s_z = \mathbf{H}_z e_x$ over $\mathbb{F}_2$, so X- and Z-type errors are decoded independently. This CSS independence is structurally important for the results of Sec. 5.2.

Bivariate bicycle codes [1] are constructed over the group algebra $\mathbb{F}_2[x,y]/(x^\ell - 1,\, y^m - 1)$. Polynomials $A(x,y)$ and $B(x,y)$ yield circulant block matrices:

$$\mathbf{H}_x = [\mathbf{A} \mid \mathbf{B}], \qquad \mathbf{H}_z = [\mathbf{B}^\top \mid \mathbf{A}^\top]. \tag{1}$$

We study three codes (Table 1).

2.2 Belief propagation and its failure modes

Min-sum BP on a CSS Tanner graph iterates variable-to-check and check-to-variable messages for T steps. For a check c_j with measured syndrome bit s_j and a neighboring variable v_i,

$$\mu^{(t)}_{v_i\to c_j} = \Lambda_i + \sum_{j'\in\mathcal{N}(i)\setminus j} \mu^{(t-1)}_{c_{j'}\to v_i}, \tag{2}$$

$$\mu^{(t)}_{c_j\to v_i} = (-1)^{s_j} \prod_{i'\in\mathcal{N}(j)\setminus i} \operatorname{sgn}\!\left(\mu^{(t)}_{v_{i'}\to c_j}\right) \times \min_{i'\in\mathcal{N}(j)\setminus i} \left|\mu^{(t)}_{v_{i'}\to c_j}\right|. \tag{3}$$

CSS decoding separates the X and Z components, processing each with its respective parity-check matrix. Under the noise model of Sec. 2.3 the matched channel LLRs are $\Lambda_{z,i} = \ln((1-p_z)/p_z)$ for Z errors and $\Lambda_{x,i} = \ln((1-p_x)/p_x)$ for X errors; at $p = 0.04$ and $\eta = 20$, $p_z = 0.0381$, $p_x = 0.0019$ and $\Lambda_x/\Lambda_z = 6.26/3.23 \approx 1.94$. Each experiment below states which prior every decoder receives.

The Tanner graphs of the finite-length BB codes considered here contain many short cycles. Message reuse around these cycles creates statistical dependencies that are absent on a tree and can lead BP to oscillate, converge to a syndrome-inconsistent estimate, or converge to the wrong logical class. We therefore treat short-cycle structure as a finite-length decoder mechanism to be measured, rather than attributing every cycle directly to the CSS orthogonality condition. Post-processing methods including ordered-statistics decoding (OSD) [11] recover many such failures but add $O(n^3)$ worst-case complexity. As shown in Sec. 5.1, the specific OSD *configuration*—schedule, base method, and iteration count—can change LER by more than two orders of magnitude on the same code. We therefore report these settings explicitly in the comparisons below.

2.3 Noise models

We use a Z-biased CSS component-noise model with $\eta = 20$:

$$p_z = \frac{p\eta}{\eta+1}, \qquad p_x = \frac{p}{\eta+1}. \tag{4}$$

Each data qubit independently suffers a Z flip with probability p_z and an X flip with probability p_x. No separate Y channel is sampled: a qubit carries a Y error only if it receives both flips, which happens with probability $p_x p_z \approx p^2/22$, so this is not depolarizing noise. Syndromes are $s_x = \mathbf{H}_x e_z$ and $s_z = \mathbf{H}_z e_x$, and the two components are decoded separately. Two kinds of temporal variation appear, both motivated by fluctuating two-level systems [29, 30]: the GNN training data modulate the global rate sinusoidally, $p(t) = p_0 + A\sin(2\pi t/T)$ (Sec. 3.2), and the drift experiment of Sec. 5.6 lets the rate of each qubit follow its own Ornstein–Uhlenbeck process. Which noise parameters a decoder sees is stated with each experiment.

3 The AdaptiveBP Framework

This section describes the model evaluated in Sec. 5. Two further variants, FiLM conditioning and an interleaved GNN-BP pipeline, are described in Appendix A; they are not evaluated.

Table 1: Bivariate bicycle codes studied in this work, with generator polynomials from [1]; $\mathbf{H}_x$ and $\mathbf{H}_z$ follow Eq. (1) over $\mathbb{F}_2[x,y]/(x^\ell-1,\, y^m-1)$. The names "gross" and "two-gross" follow IBM's usage [3].

Code	Name	n	k	d	Rate	ℓ	m	$A(x,y)$	$B(x,y)$
$[\![72,12,6]\!]$	—	72	12	6	1/6	6	6	x^3+y+y^2	y^3+x+x^2
$[\![144,12,12]\!]$	gross	144	12	12	1/12	12	6	x^3+y+y^2	y^3+x+x^2
$[\![288,12,18]\!]$	two-gross	288	12	18	1/24	12	12	$x^3+y^2+y^7$	y^3+x+x^2

3.1 TannerGNN

TannerGNN operates on the unified CSS Tanner graph, with edges distinguished by type $\tau \in \{X\text{-check}, Z\text{-check}\}$. Each node receives a 4-dimensional feature vector whose first entry is a value and whose last three entries one-hot encode the node type: a variable node v_i receives $\mathbf{x}_i = [\ell, 1, 0, 0]$, and an X-check (Z-check) node c_j receives $[s_j, 0, 1, 0]$ ($[s_j, 0, 0, 1]$), where s_j is its syndrome bit. The prior LLR $\ell = \ln((1-p)/p)$ is computed from the total physical error rate p of the shot, not from p_z or p_x: for the drifting training streams it is the instantaneous rate $p(t)$, and for the test sets the fixed p. The same ℓ is the channel prior of both CSS components in the flooding-BP decoders of Sec. 5. Its magnitude does not change min-sum decisions (Sec. 5.2), but it sets the scale against which the correction Δ_i acts.

An input projection maps features to hidden dimension d_h ($d_h = 32$ for the model of Sec. 5.3) via $\mathbf{h}_i^{(0)} = \mathrm{ReLU}(\mathrm{LayerNorm}(\mathbf{W}_{\mathrm{proj}}\mathbf{x}_i))$. Each of $L = 3$ message-passing layers applies separate edge-type MLPs, optional attention weighting, and a GRU node update:

$$\mathbf{m}_i^{(\ell)} = \sum_{j \in \mathcal{N}(i)} \mathrm{MLP}_{e,\tau(i,j)}^{(\ell)}([\mathbf{h}_i^{(\ell)}, \mathbf{h}_j^{(\ell)}]), \quad (5)$$

$$\mathbf{h}_i^{(\ell+1)} = \mathrm{GRU}(\mathbf{h}_i^{(\ell)},\, \mathbf{m}_i^{(\ell)}). \quad (6)$$

A readout MLP maps each variable node's final hidden state to a per-qubit LLR correction: $\Delta_i = \mathrm{MLP}_{\mathrm{read}}(\mathbf{h}_i^{(L)})$.

3.2 Training

The model is trained as a per-qubit classifier of the sampled Z error; BP is not part of its loss. For a shot with prior LLR ℓ on every qubit, the GNN outputs a correction Δ_j for each qubit j, giving the corrected error probability $q_j = \sigma(-(\ell+\Delta_j))$, and the loss is the focal loss [31] against the true Z error e_z,

$$\mathcal{L} = -\frac{1}{Bn}\sum_{b=1}^{B}\sum_{j=1}^{n} \alpha_t\,(1-q_t)^\gamma \log q_t, \quad (7)$$

where $q_t = q_j$ and $\alpha_t = \alpha$ if $e_{z,j} = 1$, and $q_t = 1-q_j$ and $\alpha_t = 1-\alpha$ otherwise, with $\alpha = 0.25$ and $\gamma = 2$; no syndrome-consistency term is used. The computation graph is therefore $(\ell, \mathbf{s}) \to$ TannerGNN $\to \Delta \to q \to \mathcal{L}$, and gradients reach the GNN without passing through BP. Ten-iteration flooding BP enters training only

Table 2: Architecture and training configuration of the TannerGNN model of Secs. 5.3 and 5.4.

GNN architecture	
Layers L / width d_h	3 / 32
Edge types	2 ($\mathbf{H}_x$, $\mathbf{H}_z$)
Node update	GRU, residual, layer norm
Node feature dim.	4
FiLM / attention	not used
Parameters	39,329
Training	
Data	8,000 shots, sinusoidal drift, $p \in [0.005, 0.045]$
Prior feature ℓ	$\ln((1-p)/p)$, true p of the shot, both components
Loss	focal, $(\alpha,\gamma) = (0.25, 2.0)$, on $q_j = \sigma(-(\ell+\Delta_j))$ vs. e_z; no syndrome term
Optimizer	AdamW, lr 2×10^{-3}, weight decay 10^{-4}
Batch size / epochs	64 / 20, gradient clipping 1.0
Checkpoint	fewest failures on 4,000 validation shots
Training seeds	5
BP for model selection	flooding min-sum, scaling 0.8, 10 iter.; not in the loss
Reference BP-OSD baseline	
Implementation	`ldpc` 2.4.1 `BpOsdDecoder`
`bp_method`	`ms` (min-sum)
`ms_scaling_factor`	0.8
`schedule`	`serial` (layered)
`max_iter`	100
OSD (where used)	`osd_cs`, order 10

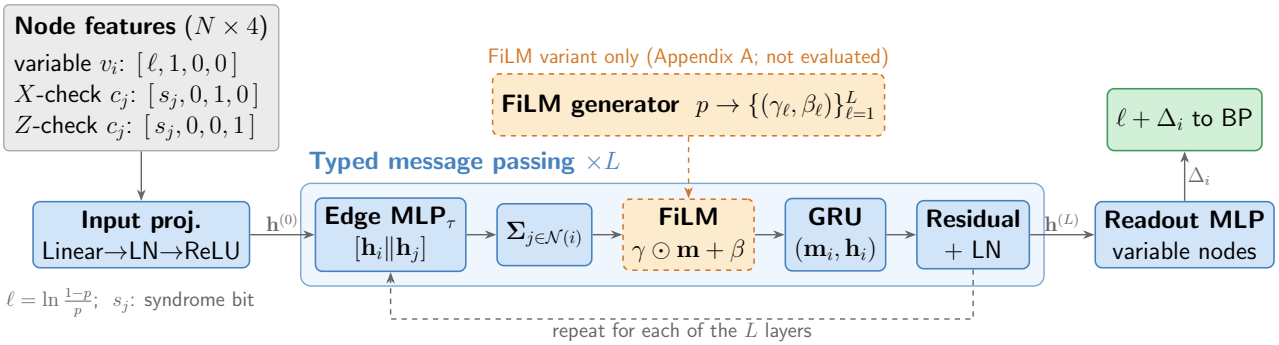


Figure 1: TannerGNN architecture. Variable and check nodes exchange typed messages over $L = 3$ layers via edge-type-specific MLPs; a GRU update with a residual connection and layer normalisation follows at each node, and a readout MLP on the variable nodes gives the LLR corrections. In the FiLM variant (dashed; Appendix A), FiLM modulation scales the aggregated messages at each layer; the model of Sec. 5.3 does not use FiLM.

through model selection: after each of the 20 epochs the corrected priors $\ell + \Delta$ are decoded with that BP on 4,000 validation shots, and the epoch with the fewest failures is kept. The training data are 8,000 shots from two sinusoidally drifting streams, $p(t) = p_0 + 0.015 \sin(2\pi t/500)$ with $p_0 \in \{0.02, 0.03\}$, so the training rates span $p \in [0.005, 0.045]$. Table 2 summarises the configuration.

The target is the sampled error e_z, so the loss is not degeneracy-aware: an estimate that differs from e_z by a stabilizer is logically correct but is penalized like any other error. The extreme case is an error that covers exactly half of a weight-six stabilizer g. Then e_z and $e_z \oplus g$ have the same weight and the same syndrome, so they are equally likely and the network receives the same input for both; the loss rewards treating the six qubits alike and gives no incentive to prefer either half. Section 5.4 shows that such configurations are a major cause of BP failures. Degeneracy-aware objectives, such as that of Liu and Poulin [24], avoid this problem; we did not use one.

Remark 1 (One correction for two CSS components)**.** *The GNN outputs one correction Δ_i per qubit and is trained on Z errors only. In the experiments of Sec. 5 the same Δ_i is added to the prior ℓ of both components, although the two error types differ: X errors are $\eta = 20$ times rarer, and the matched LLRs differ by a factor $\Lambda_x/\Lambda_z \approx 1.94$ at $p = 0.04$. Section 5.3 measures the consequence and shows that adding Δ_i to the Z component only is better.*

4 Experimental Setup

4.1 Evaluation regimes

Two distinct evaluation regimes are used throughout this paper, and the distinction is central to interpreting the results:

Regime A (development baseline): The GNN is evaluated with flooding BP, the decoder used to select its checkpoint. This comparison isolates the gain relative to that decoder.

Regime B (stronger reference decoder): The same learned correction is evaluated with serial-schedule min-sum BP-OSD (Table 2). This tests whether the correction also improves a stronger classical decoder; BP-OSD serves here as a reference point, not as the best available decoder (Sec. 1.1). Plain BP-OSD uses a uniform prior in each component (p_z and p_x, or p in Table 6); the value of a uniform prior does not change its decisions, which we confirmed on all 20,000 test shots at $p = 0.04$. With the GNN, qubit i enters BP-OSD with prior error probability $\sigma(-(\ell + \Delta_i))$ in both components. Section 5.4 shows that the correction does not help in the tested regime.

Unless otherwise stated, all experiments use Z-biased noise with $\eta = 20$.

4.2 Statistical protocol

Logical failure. For each shot let $\hat{e}_z$ and $\hat{e}_x$ be the decoder's hard decisions and $r_z = e_z \oplus \hat{e}_z$, $r_x = e_x \oplus \hat{e}_x$ the residual errors. A shot is a logical failure if (i) the estimate does not reproduce the measured syndrome, $\mathbf{H}_x \hat{e}_z \neq s_x$ or $\mathbf{H}_z \hat{e}_x \neq s_z$, or (ii) the syndrome is reproduced but the residual acts nontrivially on the code space, $\mathbf{L}_x r_z \neq \mathbf{0}$ or

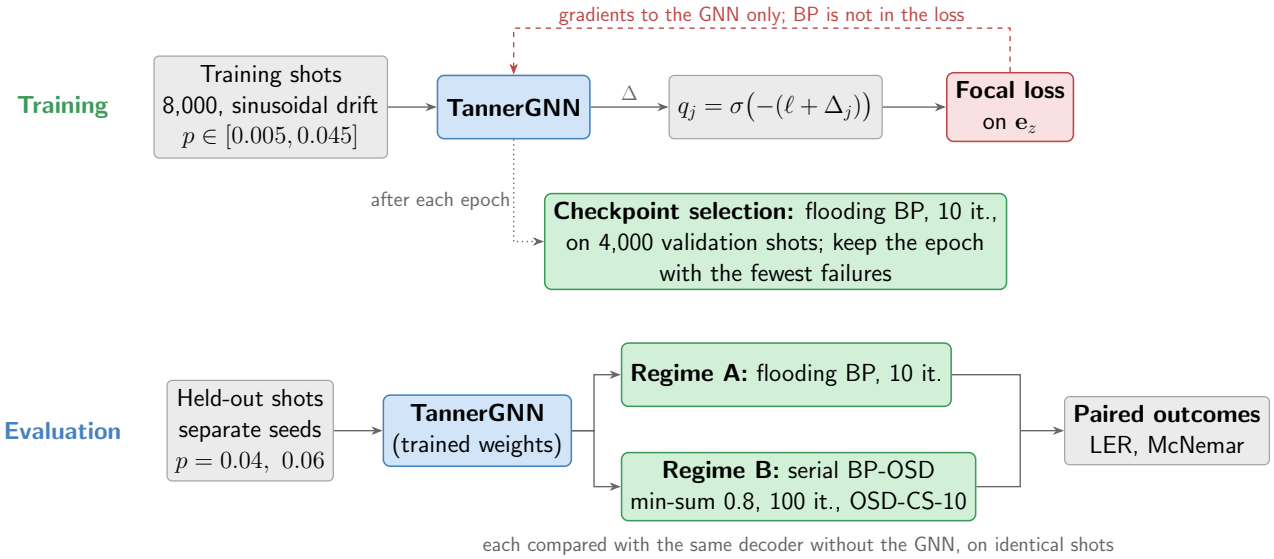


Figure 2: Training and evaluation pipeline of the GNN experiments. The GNN is trained with a per-qubit focal loss, without BP in the loss; 10-iteration flooding BP selects the checkpoint on validation shots. The model is evaluated on separately seeded held-out shots in Regime A (flooding BP) and Regime B (serial BP-OSD); each decoder receives identical syndrome batches to enable McNemar paired testing.

$\mathbf{L}_z r_x \neq \mathbf{0}$ over $\mathbb{F}_2$, where the rows of $\mathbf{L}_x$ ($\mathbf{L}_z$) span the $k = 12$ independent logical operators that detect Z (X) errors. Condition (i) is needed because a syndrome-inconsistent residual is not a logical operator, so its parity against fixed logical representatives is not meaningful; non-converged BP outputs therefore always count as failures. The X and Z components are combined by logical OR, so the reported LER is the probability that at least one of the 12 logical qubits suffers a logical error of any Pauli type in a shot. At circuit level the same rule is applied to the detector error model: a shot fails if the decoded fault set does not reproduce the detection events, or if it does and predicts the wrong observable flips.

For absolute LER comparisons we treat operating points with fewer than 100 observed logical-error events as preliminary; this is a reporting heuristic rather than a formal power calculation. Decoder pairs are evaluated on *identical* syndrome samples, enabling paired McNemar tests [32] (two-sided, with continuity correction). We report Wilson score 95% confidence intervals for LER estimates.

Comparison families. Each results table defines one family of paired tests: the ten GNN-BP vs. BP tests of Table 4 (two error rates, five seeds), the twenty GNN+OSD vs. BP-OSD tests of Table 6 (four error rates, five seeds), the nine oracle cells of Table 9, and the GNN comparisons of the drift experiment (Table 10). Within a family we control the familywise error rate with Holm's step-down procedure [33] at level 0.05 and say so where adjusted results are reported; other p-values are unadjusted and exploratory. For emphasized comparisons we report the discordant counts n_{10} and n_{01}, and we never round a p-value to zero.

Training, validation and test data are drawn with distinct random seeds, so no evaluation shot shares a random stream with a training shot, and learned-decoder results are reported over five training seeds (initialisation and minibatch order). All training and evaluation used CPU only; no GPU was available for this work. One training epoch of the 39k-parameter model of Sec. 5.3 takes about 16 s.

5 Results

5.1 Decoder configuration sensitivity

Before presenting the GNN results, we quantify the sensitivity of the classical baseline to decoder configuration. This sensitivity sets the scale against which the learned-decoder results must be interpreted.

Table 3 illustrates how strongly decoder configuration changes the finite-length result. The orig-

Table 3: Effect of BP-OSD configuration on $[\![288, 12, 18]\!]$, $p = 0.04$, $\eta = 20$, 100,000 identical shots, priors p_z and p_x of Eq. (4). 95% Wilson intervals in brackets.

Config	Schedule	Method	Iter	LER	Errors
Original BP	parallel	prod-sum	30	0.0569 [0.0555,0.0583]	5,689
Reference BP	serial	ms(0.8)	100	0.0017 [0.0015,0.0020]	174
Original BP-OSD	parallel	prod-sum	30	0.0398 [0.0386,0.0411]	3,983
Ref. BP-OSD	serial	ms(0.8)	100	0.00034 [0.00024,0.00048]	34

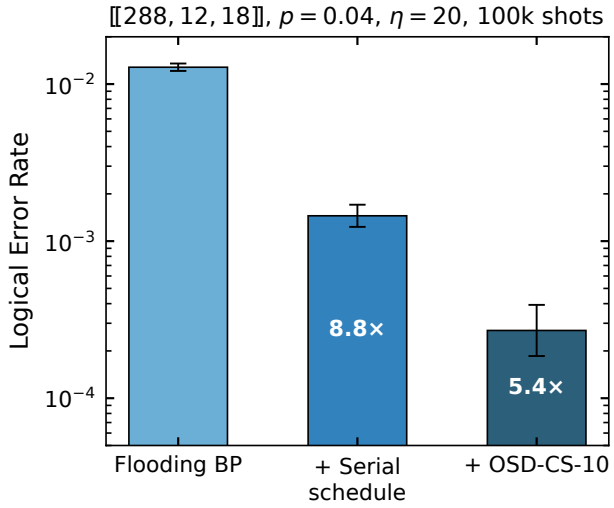


Figure 3: Baseline decomposition on $[\![288, 12, 18]\!]$, $p = 0.04$, $\eta = 20$, 100k identical shots. Min-sum BP (scaling 0.8, at most 100 iterations) with flooding and then serial schedule, then serial BP-OSD-CS-10. Wilson 95% CI error bars; labels give the factor gained at each step.

inal parallel product-sum BP-OSD configuration reduces LER from 0.0569 to 0.0398 relative to the corresponding plain BP, but remains far worse than the serial min-sum configurations. The reference serial min-sum BP-OSD records 34 logical errors in 10^5 shots, so its value is preliminary by the criterion of Sec. 4.2.

A separate $N = 100{,}000$-shot experiment (Fig. 3) changes one factor at a time. The flooding comparator is min-sum BP with scaling factor 0.8, at most 100 iterations, stopping at the first iteration whose hard decision satisfies the syndrome, with a parallel (flooding) schedule; the second decoder is identical except for the serial schedule; the third adds OSD-CS-10. The LER falls from 1.28×10^{-2} to 1.45×10^{-3} (8.8×, paired $n_{10}/n_{01} = 1138/3$) and then to 2.7×10^{-4} (5.4×, 118/0), about 47× overall.

These data do not identify a universally optimal configuration. They do show that schedule, update rule, iteration budget, and post-processing can dominate the measured finite-length LER. Learned-decoder comparisons must therefore specify and control these parameters.

5.2 Uniform-prior scaling invariance of min-sum BP

Proposition 1 (Uniform positive-scaling invariance)**.** *Consider separate-CSS min-sum BP with no message clipping or offset. If all channel LLRs in one CSS component are multiplied by the same factor $a > 0$, then every BP message and posterior LLR is multiplied by a at every iteration. Consequently, all hard decisions are unchanged.*

Proof. At initialization, each variable-to-check message is proportional to the corresponding channel LLR and therefore scales by a. Assume all variable-to-check messages at iteration t have scaled by a. The syndrome factor $(-1)^{s_j}$ in Eq. (3) is unchanged, the sign of every message is unchanged because $a > 0$, and $\min_i a|x_i| = a \min_i |x_i|$. Hence every check-to-variable message also scales by a. Equation (2) then shows that the next variable-to-check messages scale by a. Induction establishes the claim for all iterations and for the final posterior LLRs. □

For a uniform channel prior, replacing a common positive LLR Λ by another positive value Λ' is equivalent to the scaling $a = \Lambda'/\Lambda$. The proposition explains why changing only the assumed uniform physical error rate cannot change hard decisions of ideal min-sum BP. Clipping, offsets, damping rules that depend on magnitude, non-uniform priors, or sum-product updates can break the exact scaling argument.

We checked the proposition on 2,000 shots each. With unclipped min-sum BP (serial or flooding, 100 iterations), changing the assumed uniform rate from $p = 0.015$ to $p = 0.0286$ on $[\![288, 12, 18]\!]$, or from $p = 0.04$ to $p = 0.02$ on $[\![72, 12, 6]\!]$, leaves every hard decision unchanged

Table 4: GNN-BP vs. flooding BP on $[\![72, 12, 6]\!]$, $\eta = 20$, 20,000 ($p = 0.04$) and 10,000 ($p = 0.06$) held-out test syndromes, five training seeds (mean $\pm$ s.d.). n_{10} (n_{01}): shots only BP (only GNN-BP) fails. BP is flooding min-sum run for exactly 10 or 100 iterations; the last hard decision is used.

	$p = 0.04$	$p = 0.06$
BP, 10 iter.	0.1078	0.2867
GNN-BP, 10 iter.	0.0942 ± 0.0006	0.2564 ± 0.0019
relative reduction	$12.5\% \pm 0.6\%$	$10.6\% \pm 0.7\%$
n_{10} / n_{01}	313–370 / 64–95	343–394 / 66–80
BP, 100 iter.	0.0864	0.2403
Serial BP-OSD	0.0707	0.2138

Table 5: Failures by CSS component on $[\![72, 12, 6]\!]$, $p = 0.04$, $\eta = 20$, 20,000 held-out shots (counts). Z: Z errors decoded on $\mathbf{H}_x$; X: X errors decoded on $\mathbf{H}_z$; Either: the failure criterion of Sec. 4.2. GNN rows give the range over five training seeds.

Decoder	Z	X	Either
BP, 10 iter.	2,155	0	2,155
GNN-BP, Δ on both	1,845–1,899	7–54	1,874–1,906
GNN-BP, Δ on Z only	1,845–1,899	0	1,845–1,899
BP, 100 iter.	1,727	0	1,727
Serial BP-OSD	1,414	0	1,414
GNN + BP-OSD	1,439–1,451	0–1	1,439–1,451

(2,000/2,000). Two cases fall outside the proposition, and the invariance fails there: our PyTorch min-sum implementation clamps messages to ± 20 and changes 18 of 2,000 decisions on $[\![288, 12, 18]\!]$, and sum-product BP changes 25–43 of 2,000.

Remark 2 (What the proposition does and does not imply)**.** *The result applies to the BP pathway when adaptivity enters only by changing a uniform channel-prior magnitude. It does* not *establish that a global conditioning scalar is information-theoretically useless to a GNN: a learned model could combine a global noise estimate with local syndrome structure and change its nonlinear decision rule. The experiments below test a narrower hypothesis: spatially resolved per-qubit calibration can change the relative reliabilities seen by the decoder, whereas a single graph-wide scalar cannot provide that spatial information.*

5.3 GNN improvement over flooding BP

We evaluated the GNN with flooding BP—the decoder used to select its checkpoint—on separately seeded held-out syndromes: 4,000 validation shots at $p = 0.04$ select the checkpoint over 20 epochs, and test sets of 20,000 shots at $p = 0.04$ (inside the training range) and 10,000 shots at $p = 0.06$ (outside it) are each scored once on that checkpoint. Training is repeated with five seeds.

Table 4 gives the primary learned-decoder result. Over five training seeds, GNN-BP reduces LER relative to flooding BP by $12.5\% \pm 0.6\%$ at $p = 0.04$ and by $10.6\% \pm 0.7\%$ at $p = 0.06$, outside the training range (McNemar $p < 10^{-36}$ for every seed, also after Holm correction over the ten tests). The gain comes mostly from convergence: shots on which BP does not reproduce the syndrome fall from 1,740 to 838–975 of 20,000. Flooding BP with 100 iterations and serial BP-OSD both remain better than GNN-BP. A gradient audit on the five checkpoints found non-zero signal through the network (input-to-readout gradient ratio 0.63–2.19) and non-trivial LLR corrections (mean $|\Delta_i| = 0.54$–1.01).

The two CSS components. Table 5 separates the failures at $p = 0.04$ by component. All failures of BP, 100-iteration BP and BP-OSD are in the Z component: at $p_x = 0.0019$ none of them fails on the X component in 20,000 shots. The whole GNN gain is therefore in the Z component. Adding Δ to the X component as well, as in Table 4, gains nothing there and creates 7–54 X-component failures. Adding it to the Z component only removes them and gives a reduction of $13.3\% \pm 0.9\%$ ($10.8\% \pm 0.7\%$ at $p = 0.06$). This variant was chosen after seeing the test results, so we keep the pre-specified setting in Table 4. Because BP makes no X-component errors at these operating points, a separately trained X head would have nothing to correct, and leaving the X prior unchanged is the appropriate choice. With BP-OSD the X component is essentially unaffected (at most one extra failure at $p = 0.04$), so the comparisons below do not depend on this choice.

A fixed-dataset capacity test provides an additional diagnostic: a GNN trained on 1,000 fixed syndromes for 50 epochs reduced the failure count on those same syndromes from 105 to 93 (11%; McNemar $p = 0.0095$, $n_{10} = 15$, $n_{01} = 3$). This experiment tests the ability of the architecture and optimizer to fit the supplied examples; it is not a generalization result. The FiLM and inter-

Table 6: GNN+OSD vs. serial BP-OSD on $[\![72, 12, 6]\!]$, $\eta = 20$, 20,000 identical shots per p, five training seeds (ranges over seeds). n_{10} (n_{01}): shots only BP-OSD (only GNN+OSD) fails. Last two columns: seeds for which GNN+OSD is significantly worse (McNemar $p < 0.05$), unadjusted and after Holm correction over the 20 tests. No seed is significantly better.

p	BP-OSD	GNN+OSD	n_{10} / n_{01}	Seeds worse unadj.	Seeds worse Holm
0.020	0.0093	0.0095–0.0102	3 / 7–20	1/5	1/5
0.030	0.0297	0.0306–0.0317	21–28 / 46–64	5/5	4/5
0.040	0.0676	0.0688–0.0709	52–62 / 84–120	4/5	2/5
0.050	0.1354	0.1384–0.1418	94–121 / 169–222	5/5	5/5

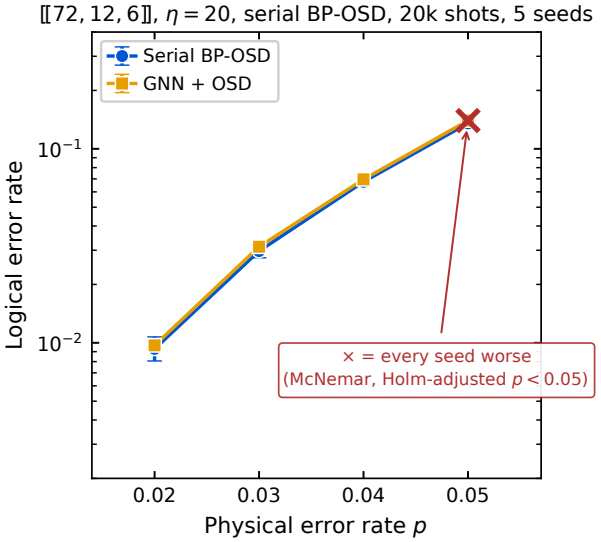


Figure 4: GNN+OSD vs. serial BP-OSD LER on $[\![72, 12, 6]\!]$, $\eta = 20$, 20,000 shots per point. BP-OSD: Wilson 95% intervals; GNN+OSD: mean over five training seeds, error bars give the range over seeds. The red × marks the operating point where every seed is significantly worse after Holm correction.

leaved variants (Appendix A) are not evaluated in this work.

5.4 Development/evaluation baseline mismatch

Although the GNN improves flooding BP, the same correction does not improve the serial min-sum BP-OSD reference configuration. Note that Tables 4 and 6 report different evaluation regimes (Regime A and Regime B of Sec. 4.1) on different syndrome sets, and their LER values are not directly comparable.

Table 6 and Fig. 4 show that the GNN-corrected decoder never improves on serial BP-OSD: no seed is significantly better at any error rate. Without correction, one seed of five is significantly worse at $p = 0.02$ and 14 of the 15 seed–error-rate pairs are significantly *worse* from $p = 0.03$ on; after Holm correction over the 20 tests, 11 of these 15 remain, including all five seeds at $p = 0.05$. The same checkpoints improve flooding BP, so these data show that the learned correction does not transfer unchanged to the stronger decoder.

Table 7: Paired failure sets on $[\![72, 12, 6]\!]$, $p = 0.04$, $\eta = 20$, 20,000 identical held-out shots. BP: 10-iteration flooding BP. GNN rows give the range over five training seeds. "Non-conv.": the estimate does not reproduce the syndrome.

Decoder	Failures	Non-conv.
BP, 10 iter.	2,155	1,740
BP, 100 iter.	1,727	700
Serial BP-OSD	1,414	0
GNN + BP, 10 iter.	1,874–1,906	838–975
GNN + BP-OSD	1,439–1,451	0
BP vs. BP-OSD failure sets		
Both fail	1,319	
Only BP fails / only BP-OSD fails	836 / 95	
BP fails, given BP-OSD fails	93%	
BP-OSD fails, given BP fails	61%	
BP-100 failures also BP failures	1,727 / 1,727	
What the GNN changes		
Shots GNN fixes for BP	313–370	
of which BP did not converge	91–93%	
of which BP-OSD also fails	31–36%	
BP-OSD failures also GNN+BP failures	87–88%	
GNN+BP-OSD vs. BP-OSD: newly failed / rescued	237–282 / 206–254	

The GNN checkpoint was selected for 10-iteration flooding BP and is then inserted ahead of a decoder with a different schedule, iteration budget, and OSD post-processing.

A paired failure-set analysis (Table 7) shows that the failure sets of BP and BP-OSD overlap strongly but are not nested: 1,319 shots fail under both, 836 only under BP and 95 only under BP-OSD, so 93% of BP-OSD's failures are BP failures and 61% of BP's failures are BP-OSD failures. The 100-iteration BP failures, by contrast, are a strict subset of the 10-iteration ones. What separates BP from BP-OSD is mainly convergence: 81% of BP failures are shots on which BP does not reach a syndrome-consistent estimate, whereas BP-OSD always returns one. Of

Table 8: Classes of the BP failures on $[\![72, 12, 6]\!]$ at $p = 0.04$ (10-iteration flooding BP, traced for 100 iterations) and of the shots the GNN repairs (range over five seeds). "Met": the hard decision reproduces the syndrome. A symmetric configuration is a Z error covering three qubits of a weight-six Z stabilizer.

Class	BP failures	GNN repairs
Wrong class (syndrome met)	415	22–32
Transient (met before it. 10)	78	5–11
Slow (met by it. 100)	1,030	249–265
Oscillating (never met)	631	37–78
Stuck (never met)	1	0
Total	2,155	313–370
symmetric configuration	606	89–137

the 313–370 shots the GNN fixes for BP, 91–93% are non-converged BP shots and only 31–36% are shots that BP-OSD fails, so the GNN's gains lie mostly where BP-OSD already succeeds. When the same corrections are passed to BP-OSD they reshuffle its failures (237–282 newly failed vs. 206–254 rescued shots), almost entirely among shots on which BP itself fails. The same pattern holds at $p = 0.06$, where 96% of BP-OSD failures are BP failures (and 71% the other way). Training directly against the target decoder, or against a differentiable surrogate of its post-processing stage, remains the appropriate test of whether learned augmentation can improve BP-OSD.

Which failures the GNN repairs. To see what kind of failure the correction repairs, we traced flooding BP for 100 iterations on the $p = 0.04$ test set (Table 8). Of the 2,155 BP failures, 415 reach a syndrome-consistent estimate in the wrong logical class. The other 1,740 do not reproduce the syndrome after 10 iterations: 78 had reproduced it earlier and moved away, 1,030 reproduce it later, 631 never reproduce it and keep changing their hard decision, and one is stuck. Stopping each shot at its first syndrome-consistent iteration would change the BP failure count only from 2,155 to 2,146, so the fixed iteration budget is not a handicap of the baseline. The shots the GNN repairs are mostly slow ones (249–265 of 313–370), followed by oscillating (37–78) and wrongly converged (22–32) shots: the correction mainly speeds up convergence.

A recurring structure among the failures is a symmetric stabilizer configuration [9]: the Z error covers exactly three qubits of a weight-six Z stabilizer g, so that e_z and $e_z \oplus g$ have the same weight and syndrome and are logically equivalent. Such configurations occur in 3.5% of the test shots but in 28% of BP failures; BP fails on 87% of the shots that contain one and on 8% of the others. Error weight does not explain this. On all 57 weight-three errors of this kind, BP, 100-iteration BP and GNN-BP with every seed fail, while BP-OSD fails on none (24 of 24 and none at $p = 0.06$); on the 4,466 weight-three errors without such a configuration BP fails on 2.4%. The GNN lowers BP's failure rate on symmetric configurations from 87% to 67–74%, all of it on configurations of weight four and more; on the weight-three ones it never helps, as expected from its loss (Sec. 3.2). OSD resolves the tie by construction, which is one reason why many BP failures are not BP-OSD failures.

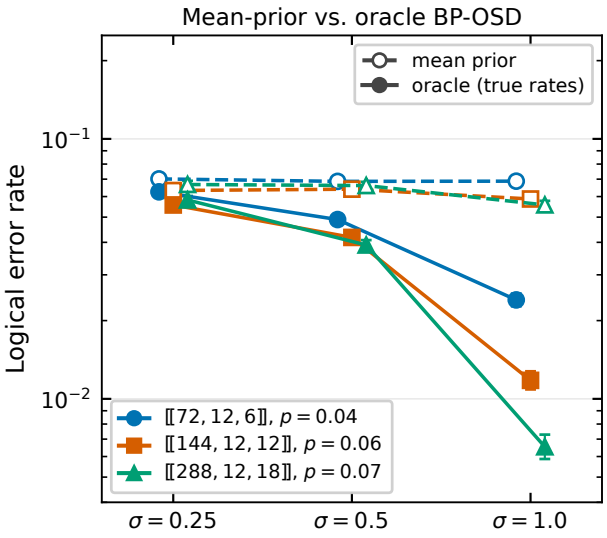


Figure 5: Mean-prior and oracle serial BP-OSD LER with Wilson 95% intervals, 50,000 identical shots per point. Each code is shown at its own operating point ($p = 0.04$, 0.06, 0.07); because code size and operating point change together, the figure does not show a distance trend.

5.5 Per-qubit calibration gap

Under spatially non-uniform per-qubit noise, each data qubit has rate $p_i = p\, e^{\sigma z_i - \sigma^2/2}$ with $z_i \sim \mathcal{N}(0, 1)$ drawn fresh every shot (capped at 0.5), so the mean rate is p for every σ; the Z and X rates of qubit i are $p_i\eta/(\eta + 1)$ and $p_i/(\eta + 1)$. A decoder knowing the true per-qubit rates (serial min-sum BP-OSD-CS-10 with the true p_i as priors) substantially outperforms a decoder using the uniform mean prior p.

Table 9: Oracle gap: LER of mean-prior and per-qubit (oracle) serial BP-OSD with Wilson 95% intervals, 50,000 identical shots per row; the three codes are at different operating points. Reduction: $1 - \mathrm{LER}_{\mathrm{oracle}}/\mathrm{LER}_{\mathrm{mean}}$. n_{10} (n_{01}): shots only the mean-prior decoder (only the oracle) fails. All nine McNemar $p < 10^{-22}$, also after Holm correction.

Code (d)	p	σ	Mean prior	Oracle	Reduction	n_{10} / n_{01}
$[\![72, 12, 6]\!]$ (6)	0.04	0.25	0.0702 [0.0680,0.0724]	0.0626 [0.0605,0.0647]	11%	681 / 301
		0.5	0.0686 [0.0665,0.0709]	0.0489 [0.0471,0.0509]	29%	1,340 / 355
		1.0	0.0688 [0.0666,0.0710]	0.0240 [0.0227,0.0254]	65%	2,435 / 198
$[\![144, 12, 12]\!]$ (12)	0.06	0.25	0.0633 [0.0612,0.0655]	0.0557 [0.0537,0.0577]	12%	924 / 543
		0.5	0.0641 [0.0619,0.0662]	0.0418 [0.0401,0.0436]	35%	1,485 / 371
		1.0	0.0587 [0.0567,0.0608]	0.0118 [0.0109,0.0128]	80%	2,441 / 94
$[\![288, 12, 18]\!]$ (18)	0.07	0.25	0.0668 [0.0646,0.0690]	0.0582 [0.0562,0.0603]	13%	1,089 / 662
		0.5	0.0661 [0.0640,0.0684]	0.0391 [0.0374,0.0408]	41%	1,804 / 450
		1.0	0.0557 [0.0537,0.0578]	0.0065 [0.0059,0.0073]	88%	2,525 / 66

Table 9 and Fig. 5 show a clear increase with noise contrast σ for each tested code. The reported percentage reduction is also larger for the larger-code rows, reaching 88% on $[\![288, 12, 18]\!]$ at $\sigma = 1.0$; however, the physical error rate is changed across codes ($p = 0.04$, 0.06, and 0.07). Consequently, these data do not isolate code distance as the cause of the cross-code trend. At a common operating point ($p = 0.04$, $\sigma = 1$) the reductions are 65%, 85% and 88% for the three codes, but the two larger codes record only 278 and 17 mean-prior failures there, so this controlled comparison is preliminary.

This experiment isolates the value of synthetic per-qubit rate side information; it does not require machine learning. We use the true-rate decoder as an *oracle reference*, not as a formal upper bound on all possible decoders. The comparison quantifies how much the tested BP-OSD implementation can benefit when the realized per-qubit rates are revealed. When the field is redrawn independently for every shot, these rates are not identifiable from that shot's syndrome: averaged over the field, the errors of each CSS component are independent across qubits with the uniform rates p_z and p_x, exactly as under uniform noise, so a component's syndrome carries no information about the realized field. (The field does correlate the X and Z errors of a qubit, which a joint decoder could exploit; ours decode the components separately.) A syndrome-only decoder could still outperform mean-prior BP-OSD by being a better decoder for i.i.d. noise, but such a gain would not come from estimating the p_i.

5.6 Drift adaptation

We test whether per-qubit rate estimates help BP-OSD when the rates drift. On $[\![72, 12, 6]\!]$ with mean rate $p = 0.04$ and $\eta = 20$, each data qubit i has a log-rate offset $x_i(t)$ that follows a discrete Ornstein–Uhlenbeck (AR(1)) process, independently across qubits,

$$x_i(t+1) = \rho\, x_i(t) + v\, \xi_i(t), \qquad \rho = \sqrt{1 - v^2/\sigma^2}, \tag{8}$$

with $\xi_i(t) \sim \mathcal{N}(0, 1)$, $x_i(0) \sim \mathcal{N}(0, \sigma^2)$ and $\sigma = 1$, so the stationary spread of the log-rates is σ for every volatility v. The volatility is the per-shot standard deviation of the log-rate innovation: $v = 0$ is a static field, and $v = 0.1$, 0.35 and 0.5 give correlation times $-1/\ln\rho$ of about 200, 15 and 7 shots. The rate of qubit i at shot t is $p_i(t) = \min(0.5,\, p\, e^{x_i(t) - \sigma^2/2})$, whose stationary mean is p, and each shot draws independent Z and X flips with probabilities $p_i\eta/(\eta+1)$ and $p_i/(\eta+1)$.

All decoders are serial min-sum BP-OSD (Table 2) with per-qubit priors $\hat{p}_i\eta/(\eta+1)$ and $\hat{p}_i/(\eta+1)$; they differ only in the rate estimate $\hat{p}_i(t)$. The *oracle* uses the true $p_i(t)$. *Stale* uses $p_i(0)$, a perfect calibration taken at the start of the stream and never updated. *Mean* uses the uniform rate p. *Empir* counts the Z flips n_i in its own decoded estimates over the previous $W = 32$ shots and shrinks the frequency toward p with $K = 16$ pseudo-shots, $\hat{p}_i = \frac{\eta+1}{\eta}\,(n_i + Kp\eta/(\eta+1))/(W+K)$. *GNN* is a TannerGNN rate estimator with the architecture of Table 2 that maps the violated-check frequencies of the previous W syndromes to an estimate $\hat{x}_i(t)$ and sets $\hat{p}_i = \min(0.5,\, p\, e^{\hat{x}_i - \sigma^2/2})$; it sees syndromes only,

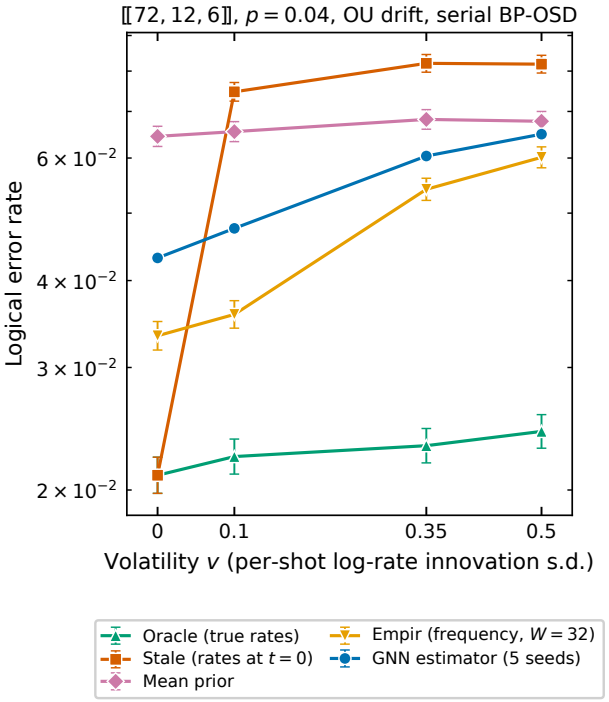


Figure 6: LER of serial BP-OSD under per-qubit OU drift against the volatility v on $[\![72, 12, 6]\!]$, $p = 0.04$, $\sigma = 1$, for five rate estimates. Wilson 95% intervals; for the GNN, the mean over five training seeds with the range over seeds.

never decoder outputs or true rates. It is trained by mean-squared error on 20,000 windows drawn at the four volatilities, with five training seeds; for each seed the epoch with the lowest validation error out of 30 is kept. Every decoder sees the current syndrome. Each volatility is evaluated on 20 independent streams of 2,532 shots, whose first 32 shots are warm-up and not scored, giving 50,000 scored shots. Shots in one stream share a rate field, so pooled McNemar tests overstate the evidence; we therefore compare decoders by a sign test over the 20 streams, with Holm correction over the eight GNN comparisons.

Table 10 and Fig. 6 show three things. First, a per-qubit calibration is valuable only while it is current. Without drift the stale calibration is the oracle (LER 0.0210, against 0.0645 for the mean prior), but at $v = 0.1$, where the rates decorrelate over about 200 shots, it is already worse than the mean prior (0.0747 vs. 0.0655). Second, tracking the rates recovers much of the gap: the non-learned frequency estimator reaches 0.0333 without drift and 0.0358 at $v = 0.1$, and its advantage shrinks as the correlation time approaches the window length (0.0601 vs. 0.0678 for the mean prior at $v = 0.5$). Third, the GNN estimator beats stale calibration at every nonzero volatility and the mean prior at every volatility (on all 20 streams up to $v = 0.35$ and on 17 of 20 at $v = 0.5$), but it is worse than the frequency estimator on all 20 streams at every volatility; all eight GNN comparisons remain significant after Holm correction. The five training seeds give nearly identical results. Under independent per-qubit drift, counting each qubit's decoded errors therefore captures more of the available signal than the learned syndrome-based estimator; whether spatially correlated drift would change this is an open question.

5.7 Circuit-level evaluation

Circuit and noise model. We simulate a Z-basis memory experiment on $[\![72, 12, 6]\!]$ with 6 rounds of syndrome extraction (72 data qubits, 36 X-check and 36 Z-check ancillas). Data qubits are prepared in $|0\rangle$; each round measures all X-checks and then all Z-checks with dedicated ancillas, using CNOT layers from a greedy conflict-free schedule; the data qubits are finally measured in the Z basis. Detectors are (i) the 36 Z-checks of round 1, which are deterministic for the prepared state, (ii) the XOR of consecutive measurements of every check in rounds 2–6, and (iii) the XOR of the last Z-check outcome with the parity of the measured data qubits in its support, giving 432 detectors and 12 logical observables. The noise, summarized in Table 11, is a custom biased Pauli model, not standard circuit-level depolarizing noise: every one- and two-qubit gate is followed by an independent single-qubit channel on each qubit it acts on, with mutually exclusive X and Z faults and no Y fault, so a two-qubit gate has no correlated two-qubit faults. Under spatially non-uniform noise each qubit q (data and ancilla) has rate $p_q = p\, e^{\sigma z_q - \sigma^2/2}$, $z_q \sim \mathcal{N}(0, 1)$, $\sigma = 1$, on all of its fault locations, so the mean rate is p as in Sec. 5.5.

Decoders. Stim [34] converts each noisy circuit into a detector error model (DEM) with 4,068 fault mechanisms on 432 detectors. Each mechanism collects the circuit faults that flip the same detectors and observables, and its prior is their combined probability; the exclusive X and Z outcomes of a channel are treated as independent faults (Stim's `approximate_disjoint_errors`).

Table 10: Drift experiment on $[\![72, 12, 6]\!]$, $p = 0.04$, $\eta = 20$, $\sigma = 1$, $W = 32$: LER of serial BP-OSD with different per-qubit rate estimates, 50,000 scored shots (20 streams) per volatility v. GNN: range over five training seeds. Last two columns: streams on which the seed-averaged GNN has fewer / more failures than stale calibration and than the empirical estimator.

v	Oracle	Stale	Mean	Empir	GNN	vs. stale	vs. empir
0.00	0.0210	0.0210	0.0645	0.0333	0.0428–0.0433	0/20	0/20
0.10	0.0223	0.0747	0.0655	0.0358	0.0474–0.0477	20/0	0/20
0.35	0.0232	0.0821	0.0682	0.0541	0.0602–0.0607	20/0	0/20
0.50	0.0243	0.0819	0.0678	0.0601	0.0648–0.0651	20/0	0/20

Table 11: Circuit noise for a qubit q with rate p_q ($p_q = p$ under uniform noise), $\eta = 20$. Each line acts independently on every qubit concerned.

Location	Fault
After reset to $\|0\rangle$	X, prob. p_q
After H or CNOT (each qubit)	X, prob. $p_q/(\eta+1)$, or Z, prob. $p_q\eta/(\eta+1)$
Idle qubit, each time step	as after a gate
Before Z measurement	outcome flip, prob. p_q

Table 12: Circuit-level oracle gap on $[\![72, 12, 6]\!]$ (6 rounds, $\eta = 20$, $\sigma = 1$). LER in percent with Wilson 95% intervals. n_{10} (n_{01}): shots only the mean-prior decoder (only the oracle) fails.

p	Shots	Mean prior	Oracle	n_{10}	n_{01}	p_{McN}
0.001	10,000	0 [0,0.04]	0 [0,0.04]	0	0	—
0.003	10,000	0 [0,0.04]	0 [0,0.04]	0	0	—
0.01	20,000	1.35 [1.19,1.51]	0.60 [0.50,0.72]	177	28	5×10^{-25}

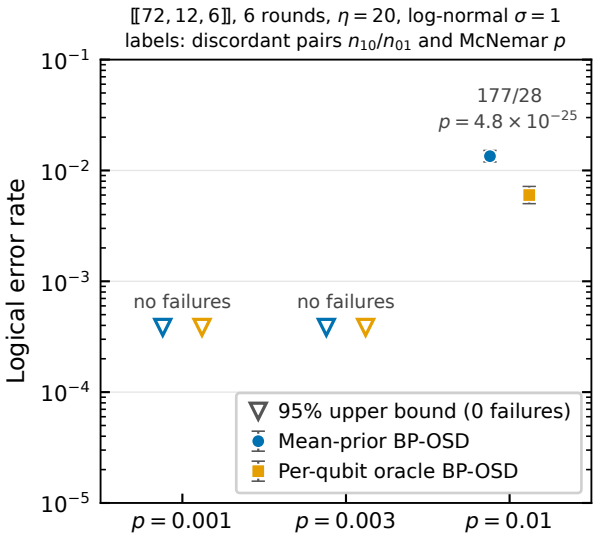


Figure 7: Circuit-level oracle gap on $[\![72, 12, 6]\!]$, 6 rounds, $\eta = 20$, $\sigma = 1$. Points: LER with 95% Wilson intervals; open triangles: 95% upper bound when no failures were observed. Labels: discordant pairs and McNemar p.

The mean-prior decoder uses the DEM of the circuit with every qubit at rate p, and the oracle uses the DEM of the circuit with each qubit at its true rate p_q; the two DEMs have the same mechanisms and differ only in their priors. Both are decoded with serial min-sum BP-OSD (scaling 0.8, 100 iterations, OSD-CS order 10), as in Table 2. We draw 20 rate profiles at $p \le 0.003$ and 40 at $p = 0.01$, with 500 shots per profile. Failures follow Sec. 4.2.

Knowing the per-qubit rates also helps at circuit level (Table 12, Fig. 7). At $p = 0.01$ the oracle reduces the logical error rate from 1.35% to 0.60% (55%; McNemar $p = 4.8 \times 10^{-25}$), and it is better on 38 of 40 independent rate profiles (sign test $p = 1.5 \times 10^{-9}$), so the result is not driven by a few extreme profiles. At $p = 0.001$ and $p = 0.003$ neither decoder fails in 10^4 shots, so these operating points cannot resolve a gap. The circuit-level gap is somewhat smaller than the code-capacity gap on the same code (65% at $\sigma = 1$, Table 9), at a different operating point, but it does not vanish.

6 Discussion

6.1 The GNN's role and its limits

TannerGNN reduces the LER of the flooding BP decoder used to select its checkpoint by 12.5%±0.6% over five training seeds, and gradient diagnostics show that the network produces nontrivial corrections. The gain comes mainly from helping BP converge on shots where it converges slowly, and 100-iteration BP and serial BP-OSD remain stronger than the corrected decoder.

The Regime-B result is also informative: the learned correction does not transfer automatically

to serial BP-OSD. This is consistent with the mismatch between the development and evaluation decoders: most BP-OSD failures are also BP failures, and the GNN's repairs fall mostly on shots BP-OSD already decodes (Table 7). The symmetric stabilizer configurations of Sec. 5.4 show the reverse case: failures that BP-OSD avoids by construction and that a bitwise loss cannot teach the GNN to repair. Training against BP-OSD directly is complicated by the non-differentiable OSD stage; differentiable surrogates, distillation, training on the residual failures of the target decoder, or a degeneracy-aware loss [24] are possible alternatives.

6.2 Benchmark design implications

The baseline experiments identify two reporting requirements for learned QLDPC decoders. First, BP-OSD schedule, update rule, iteration count, and OSD configuration must be stated explicitly because they can change LER by more than two orders of magnitude on the same code. Table 2 records these settings for the present study.

Second, the decoder used for model selection and the evaluation decoder must be distinguished. An LLR correction selected for flooding BP need not remain beneficial when the schedule, iteration count, and OSD stage are changed. Comparisons across such decoder configurations should identify the mismatch rather than treating all BP-based baselines as equivalent.

6.3 Comparison with prior work

Astra [6] uses a learned Tanner-graph message-passing decoder and reports BB-code performance competitive with, and in several settings better than, BP-OSD under code-capacity depolarizing noise. This places our 12.5% flooding-BP gain in context: improvement over plain BP is not, by itself, the main novelty. Our contribution is instead the controlled comparison showing how that gain changes when the baseline and side information are strengthened.

Gong et al. [5] interleave differentiable BP stages with GNN corrections on the same sparse graph and show error-floor improvements over several post-processing approaches. Our interleaved variant (Appendix A) follows the same broad design principle, but its decoding performance is not evaluated here, so we do not claim a competitive result for that architecture.

Recent circuit-level work on BB codes also reports neural decoders that can outperform BP-OSD on the $[\![72, 12, 6]\!]$ code while retaining a runtime advantage, although scaling to $[\![144, 12, 12]\!]$ remains more challenging [25]. This comparison reinforces the need to evaluate learned decoders against a fully specified strong non-learned baseline, not only against flooding BP.

Stein et al. [26] condition a neural QEC decoder on spatially resolved hardware calibration data using FiLM and evaluate on IBM repetition-code experiments. Their conditioning signal is per-qubit and calibration-derived, unlike the single global rate p fed to our FiLM variant (Appendix A). This distinction is consistent with the oracle experiments of Sec. 5.5, which motivate spatial rather than purely global conditioning.

6.4 Towards spatially-resolved conditioning

The oracle-gap experiment (Sec. 5.5) shows that per-qubit rate side information can materially improve the tested BP-OSD decoder. Because operating point and code size co-vary, the current data do not establish how this value scales with distance. Proposition 1 also shows that a uniform rescaling of the min-sum BP prior cannot reproduce the effect of spatially varying reliabilities. These results motivate testing spatially resolved FiLM (SRCF), in which per-qubit parameters (γ_i, β_i) are conditioned on persistent hardware calibration data rather than on a single graph-wide scalar. Soft syndrome information [35] is another form of side information that iterative decoders can use. Such a decoder should be compared with the non-learned frequency estimator of Sec. 5.6 and evaluated in a circuit-level setting where the calibration signal remains identifiable. Neither condition has been established in the present experiments.

6.5 Limitations

(i) The evaluated GNN was trained with a per-qubit loss and selected for 10-iteration flooding BP; no model was trained or selected for serial BP-OSD, our reference decoder. Whether GNN augmentation of serial BP-OSD yields gains is untested.

(ii) GNN checkpoints exist only for $[\![72, 12, 6]\!]$; the larger codes were not trained due to

CPU-only compute constraints. The null result in Regime B should not be extrapolated to the larger codes.

(iii) The interleaved and FiLM architectures (Appendix A) are described but their decoding performance is not evaluated.

(iv) The correction is trained on Z errors and, in the headline setting, also added to the X component, where it only adds failures (Table 5); the Z-only setting was chosen after seeing the test results.

(v) The training loss is not degeneracy-aware (Sec. 3.2).

(vi) Code-capacity noise omits measurement errors, leakage, and correlated faults present in real circuits, and the circuit model uses independent single-qubit channels rather than correlated two-qubit faults.

(vii) The drift experiment is synthetic: the rates of different qubits drift independently, with correlation times of 7 to 200 shots.

7 Conclusion

On $[\![72, 12, 6]\!]$, a Tanner-graph GNN whose checkpoint is selected with 10-iteration flooding BP reduces the held-out LER of that decoder by $12.5\% \pm 0.6\%$ at $p = 0.04$ over five training seeds, all of it in the Z component. The same correction never improves serial min-sum BP-OSD and is often significantly worse; it mostly repairs shots on which BP converges slowly and which BP-OSD already decodes. Weight-three errors that cover half of a stabilizer defeat BP with or without the GNN but never BP-OSD, and the bitwise training loss cannot teach the network to break that tie. The demonstrated gain is therefore specific to the flooding-BP baseline.

BP schedule, update rule, iteration budget and OSD configuration change the finite-length LER by orders of magnitude, so they belong to the decoder specification. A uniform rescaling of the channel LLRs cannot change min-sum decisions, whereas the true per-qubit rates reduce BP-OSD's LER by up to 88% under synthetic heterogeneous noise and by 55% under circuit-level noise. Under per-qubit drift a stale calibration quickly loses its value, and a simple frequency estimator tracks the rates better than a learned one. A natural next test is a decoder conditioned on persistent spatial calibration data, trained with a degeneracy-aware loss, and compared with fully specified BP-OSD and the frequency estimator.

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Data and code availability

All code, data, trained checkpoints and per-shot outcomes behind every table and figure are available at https://github.com/kondshk/adaptive_gnn (tag `paper-v5`). The script `make_tables.py` rebuilds every results table from the stored outcomes, and `requirements-lock.txt` lists the exact package versions. Code-capacity data are sampled directly with NumPy; circuit-level data are generated with Stim [34].

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A Unevaluated variants

The code base contains two further variants of the model. Neither is evaluated in this paper, and no result in the main text depends on them.

FiLM conditioning. Feature-wise Linear Modulation (FiLM) [36] conditions each GNN layer on an external context. A small generator network maps a global noise level to per-layer affine parameters $(\gamma^{(\ell)}, \beta^{(\ell)})$, which modulate the aggregated messages as $\gamma^{(\ell)} \odot \mathbf{m}^{(\ell)} + \beta^{(\ell)}$ (Fig. 1). In the implementation the generator receives the physical error rate p of the shot, i.e. the true global rate rather than an estimate from the syndrome. This input is one scalar shared by the whole graph and contains no per-qubit reliability information. Proposition 1 shows that uniformly rescaling the channel prior cannot change the hard decisions of unclipped min-sum BP; it does not imply that a GNN cannot use a global scalar as context.

Neural BP. Differentiable BP generalises Eqs. (2)–(3) with learnable per-iteration weights on the channel LLR, the variable-to-check messages and the check-to-variable messages, $w^{(t)} = \mathrm{softplus}(w_{\mathrm{raw}}^{(t)})/\ln 2$. Dividing by $\ln 2$ makes $w_{\mathrm{raw}} = 0$ give $w = 1$ exactly, so at initialisation the unrolled network is standard min-sum BP with scaling factor 0.8.

Interleaved pipeline. Algorithm 1 and Fig. 8 give the interleaved pipeline. The GNN intervenes twice: before BP to correct channel priors, and after T_1 iterations using the intermediate posterior information. Qubits with $p_i^{\mathrm{mid}} \approx 0.5$ have low posterior confidence and can receive a second learned correction. Unlike the model of Sec. 3.2, both correction stages are trained jointly through the differentiable BP unroll, so the loss is computed on BP's output marginals. The FiLM and interleaved variants are trained on data regenerated each epoch with $p \sim \mathrm{Uniform}[0.01, 0.10]$, with the focal loss on both components and a syndrome-consistency term, the binary cross-entropy between the measured syndrome and $\frac{1}{2}(1 - \prod_{j \in \mathcal{N}(c)}(1 - 2q_j))$ for each check c, with weight 0.1.

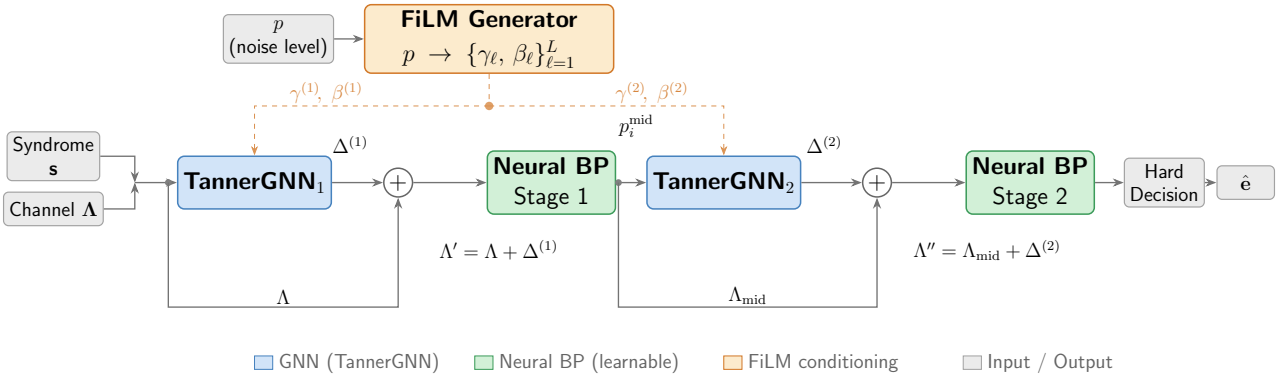


Figure 8: Interleaved GNN-BP pipeline. The GNN applies two successive LLR corrections: one before BP initialisation and one after T_1 iterations, using the mid-decoding posterior as an additional input. Both stages are trained end-to-end through the BP unroll. This variant is not evaluated in this paper.

Algorithm 1: Interleaved GNN-BP Decoding

Input: $\mathbf{s}$, channel LLR Λ, global noise level p

Output: Error estimate $\hat{\mathbf{e}}$

$(\gamma, \beta) \leftarrow \text{FiLMGen}(p)$ `// Conditioning`

$\Delta^{(1)} \leftarrow \textsc{TannerGNN}(\Lambda, \mathbf{s}, \gamma, \beta)$

$\Lambda' \leftarrow \Lambda + \Delta^{(1)}$ `// Prior correction`

$\Lambda_{\text{mid}} \leftarrow \text{NeuralBP}(\Lambda', \mathbf{H}, T_1)$

$p_i^{\text{mid}} \leftarrow \sigma(-\Lambda_{\text{mid},i})\ \forall i$

$\Delta^{(2)} \leftarrow \textsc{TannerGNN}(\Lambda_{\text{mid}}, p^{\text{mid}}, \mathbf{s}, \gamma, \beta)$

$\Lambda'' \leftarrow \Lambda_{\text{mid}} + \Delta^{(2)}$ `// Mid correction`

$\Lambda_{\text{final}} \leftarrow \text{NeuralBP}(\Lambda'', \mathbf{H}, T_2)$

$\hat{e}_i \leftarrow \mathbb{1}[\Lambda_{\text{final},i} < 0]\ \forall i$

return $\hat{\mathbf{e}}$